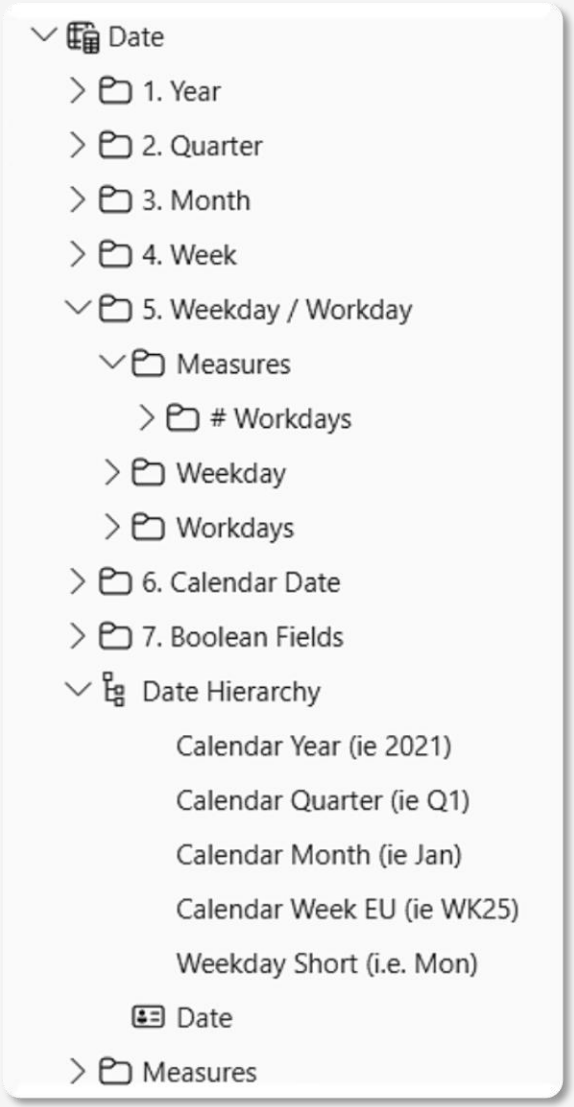
Power BI’s Hottest Upgrade - Developer Mode with TMDL



Dhyan Rathore

20 AUG 2026

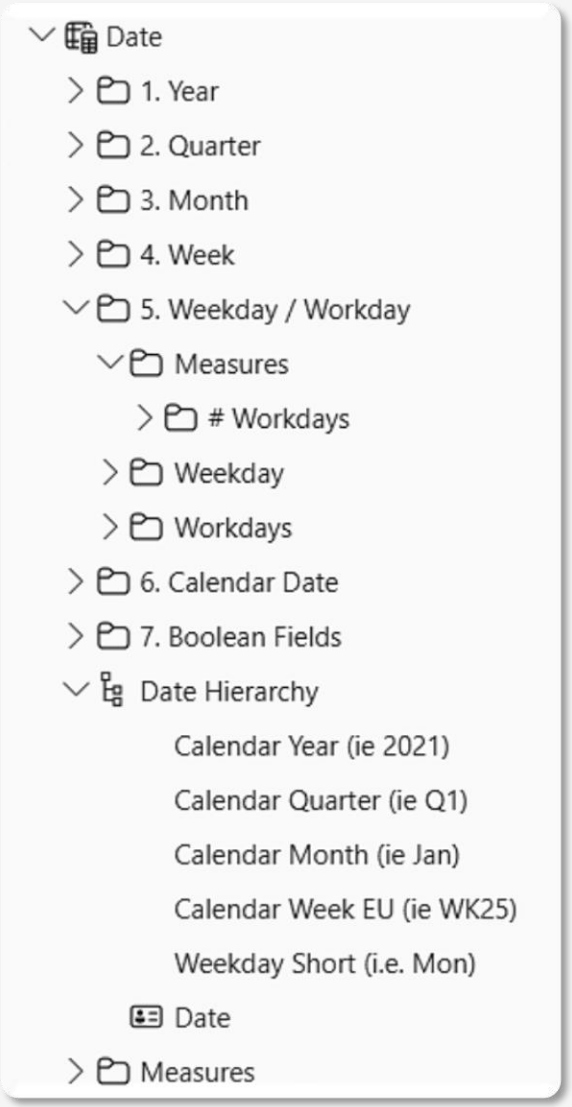


Reuse options

Create from scratch

External tools

TMDL



Reuse options

Create from scratch

40 minutes

~~***External tools***~~

TMDL

10 seconds



Dhyan Rathore

Fabric Enablement Lead @ Autoliv

- ✓ **Stockholm, Sweden**
- ✓ **Data & Mushrooms**

dhyans

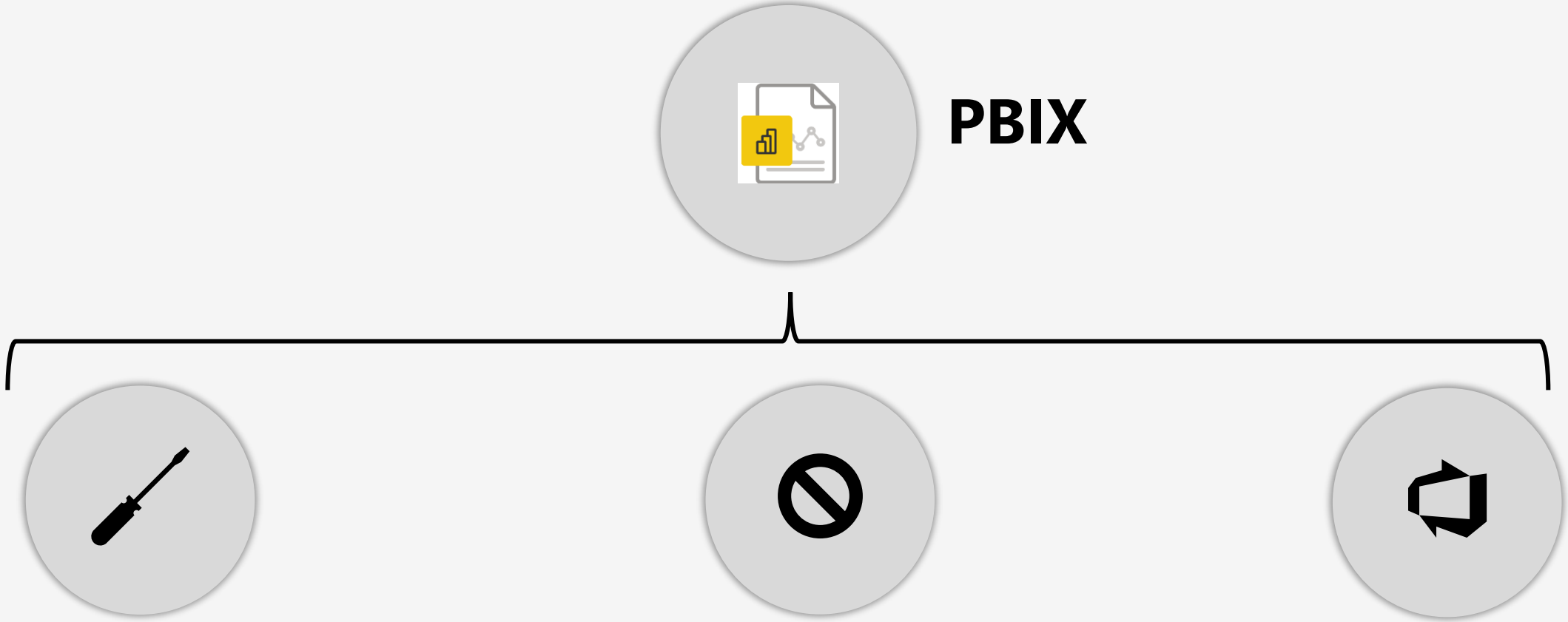
bits2BI

Agenda

- TMDL: What, Why, When & How
- Using TMDL with Local and Remote Models
- TMDL the Smart Way

TMDL…???

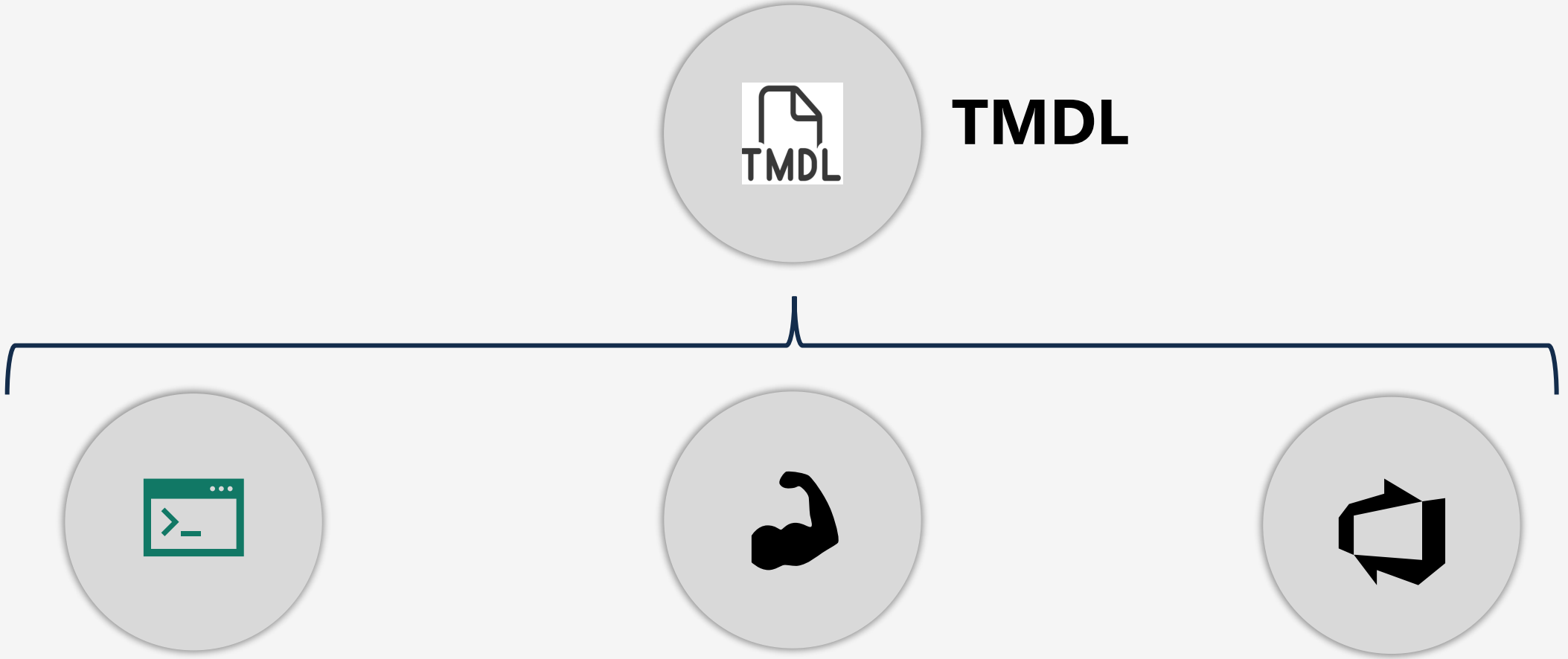
Why do we need TMDL



Version Control

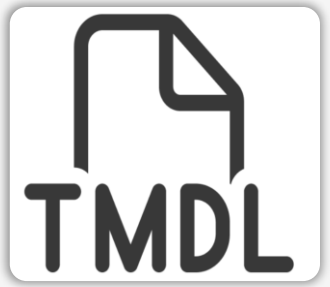


Why do we need TMDL



What is TMDL

Tabular **M**odel **D**efinition **L**anguage



- **Serialized** metadata of a semantic model
- Each model object is represented as text

Conceptualized by the Community: **Mathias Thierbach**

What can we do with TMDL



Create Objects



Edit Objects & Properties



Objects & Properties without a UI in PBI Desktop



Modeling with AI

When to use TMDL

- Viewing & editing multiple objects at once
- Work with objects & properties without an UI
- Reuse objects across different models
- View & edit Power Query transformations
- Take a “one-page” backup of semantic model
- Working with a code first experience
- Using AI for modelling and editing

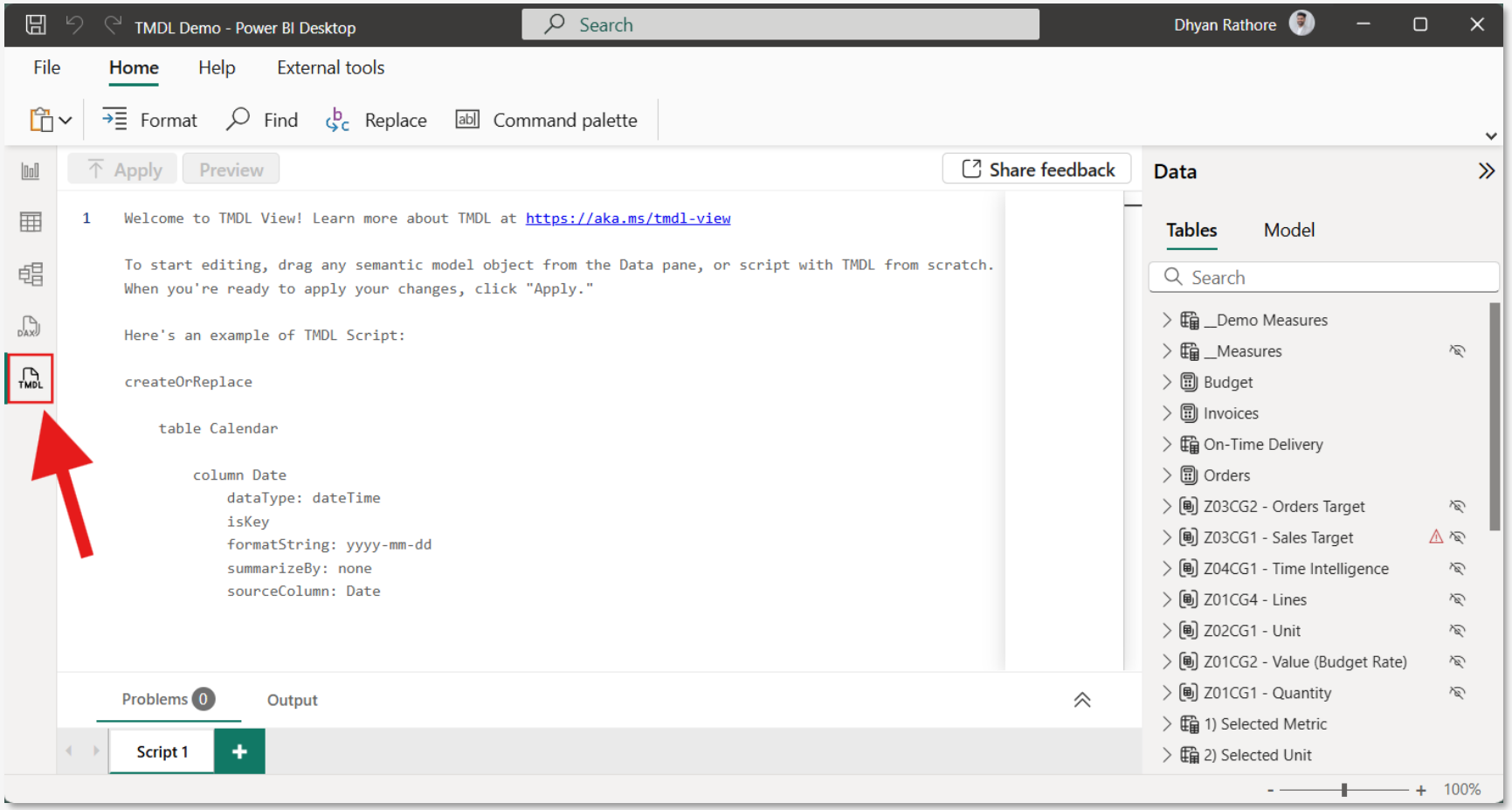
Let’s see it in Action!

Different Types of TMDL

TMDL Scripts

TMDL View

- **Power BI Desktop – GA**
- **Power BI Web – Preview**

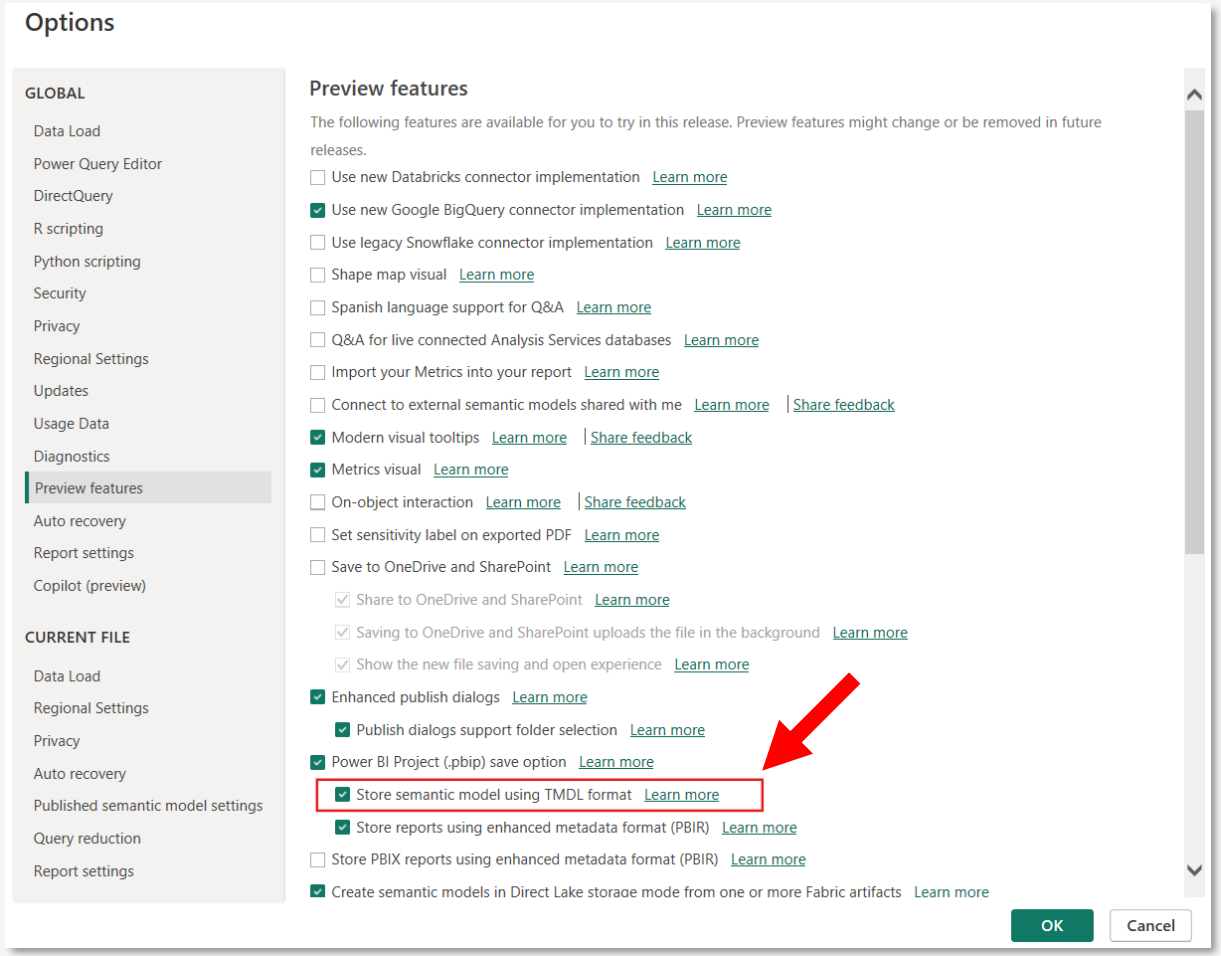


TMDL File Format

Enable TMDL File Format

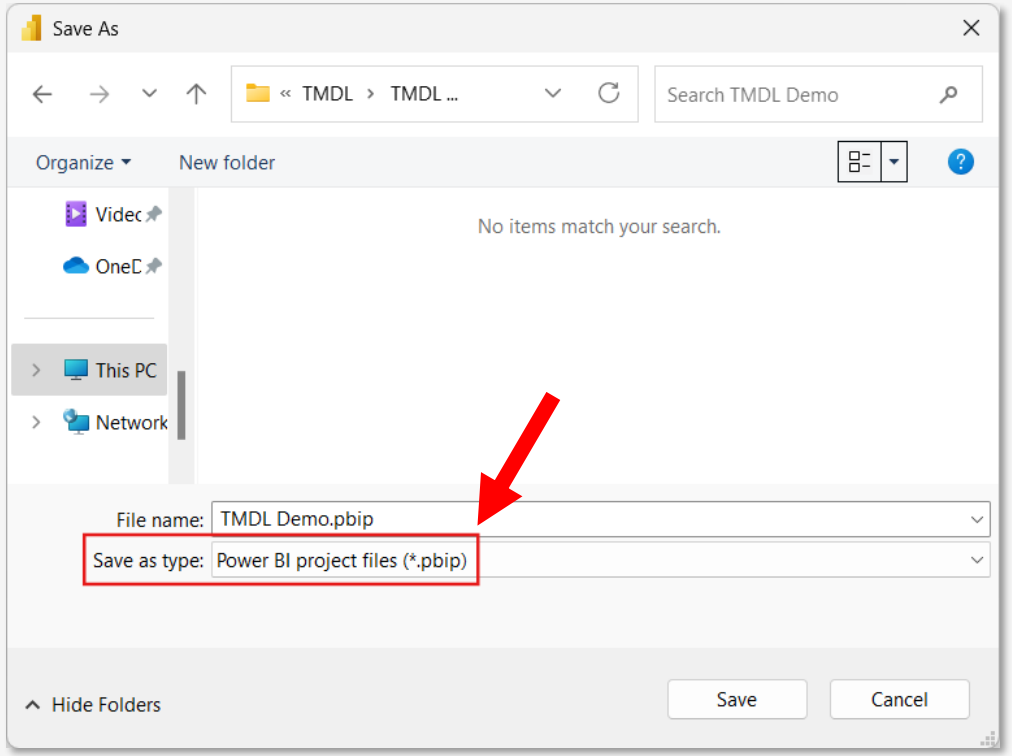
Preview in Desktop

File > Options and settings > Options > Preview features (only with pbip)



Save pbix as .pbip

File > Save as



In Power BI Service, TMDL is the default storage mode for semantic models

Summary

TMDL Scripts GA

Script containing the object metadata as text that Power BI **Desktop** can use to make changes to the model.

Keywords: *createOrReplace, ref*

```
createOrReplace

    table Date
        lineageTag: 0bb45e29-3908-422d-ade4-4100434bdfa4
        dataCategory: Time

        measure '# Workdays MTD' =
                CALCULATE(
                    MAX( 'Date'[Workdays MTD] ),
                    'Date'[IsDateInScope] = TRUE
                )
            formatString: #,##0
            displayFolder: 5. Weekday / Workday\Measures\# Workdays
            lineageTag: 4b0555ab-6912-4d3c-b613-0d2d542a6965
```

TMDL Files Preview in Desktop

Files storing object metadata as text that Power BI can use to render a model.

File extension: *tmdl*

```
table Date
    lineageTag: 0bb45e29-3908-422d-ade4-4100434bdfa4
    dataCategory: Time

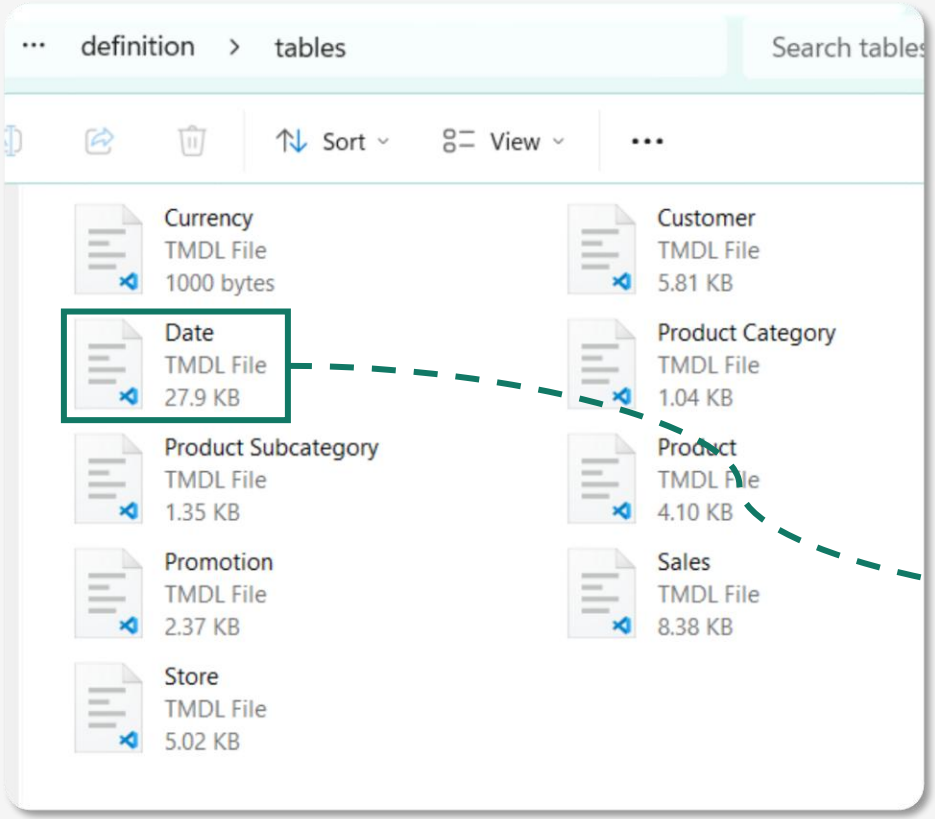
    measure '# Workdays MTD' =
            CALCULATE(
                MAX( 'Date'[Workdays MTD] ),
                'Date'[IsDateInScope] = TRUE
            )
        formatString: #,##0
        displayFolder: 5. Weekday / Workday\Measures\# Workdays
        lineageTag: 4b0555ab-6912-4d3c-b613-0d2d542a6965
```

One can be used without using the other

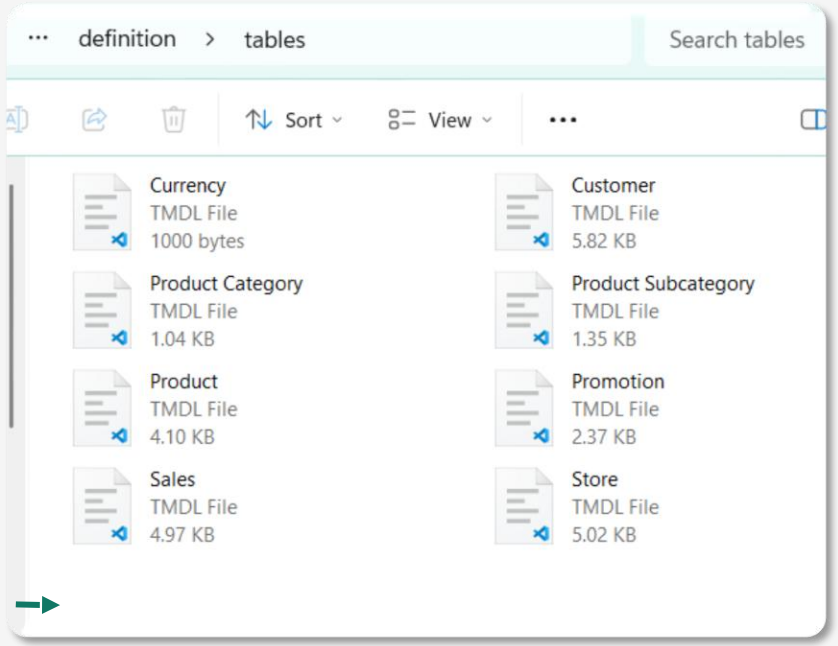
Can we use TMDL files to do things that we did with TMDL scripts?

Using TMDL Files with *Local Models*

Local Model: Reuse Objects with TMDL Files



Source Model: TMDL Files



Target Model: TMDL Files

1. Move the object TMDL files from source to target model TMDL files
2. Relaunch Power BI Desktop
3. Refresh the object (if required)

Power BI Desktop auto-detect changes: Planned

Local Model: Working with TMDL Files

Tool Up



Visual Studio Code (VS Code)



TMDL Extension for VS Code

Updating *Remote Models* with TMDL

Remote Model: Update with TMDL Scripts

TMDL View in Web – Preview

- **Everything as TMDL view in Power BI Desktop + Features only supported in the web e.g., assign RLS roles**
- **Unlike Desktop, scripts are not saved.**

Editing data models in the Power BI service must be enabled

Remote Model: Update with TMDL Files

Tool Up



Visual Studio Code (VS Code)



TMDL Extension for VS Code



Fabric Studio Extension for VS Code

TMDL with AI

TMDL with AI

- AI Agents use TMDL to create and edit semantic models.
- Use TMDL ‘one-page’ backups:
  - To track model edits by the agents
  - To use GenAI to update your models

TMDL the Smart Way

When NOT to use TMDL



Editing DAX or Power Query



When the user interface is simpler



Changing unknown properties



Creating Perspectives for the first time

Tips

- Default values of properties are not shown
- A closed TMDL script tab can’t be recovered
- Preview before applying changes
- No undo for data modeling changes

Some Warnings



Careful with Find & Replace as you can make unintended changes



Dependencies are not cross highlighted



Careful with changing properties that you don’t fully understand



TMDL only changes the model metadata and does not update the visuals

Recap



What, Why, When & How



TMDL Scripts



TMDL files



Tips & Warning

Get the slides…



https://github.com/DhyanRathore/Presentations*/tree/main/2026/20260820_SwedishPowerBI&FabricUserGroup*

Thank You!!

Dhyan Rathore

dhyans